

Solution Architecture

2. High-Level Architecture

The system has four layers.

Layer 1 — Browser. The user sees HTML pages rendered from Jinja2 templates. Each planner page has a form; JavaScript reads the form, sends it to the server with `fetch`, and shows the returned plan on the page.

Layer 2 — FastAPI application (`app.py`). Receives requests, checks input using the Pydantic models in `models.py`, checks login using `auth.py`, calls the right planner function, saves the result to history and returns it.

Layer 3 — AI utility layer (`gemini_utils.py`). Builds the prompt, calls Gemini, cleans and parses the JSON answer, and attaches shopping links.

Layer 4 — External and in-memory storage. Google Gemini is the external AI service. Users, blacklisted tokens and history are kept in Python dictionaries and sets in memory.

Request path in one line: Browser → FastAPI route → validation and login check → planner function → Gemini → JSON → shopping links → history → response to browser.

3. Module Design

`app.py` — creates the FastAPI app, enables CORS, mounts the static folder, defines all routes, saves uploaded outfit images to `static/uploads/` with a random file name, and stores every result in `history_db`.

`auth.py` — holds `users_db` and `blacklisted_tokens`; provides `hash_password`, `verify_password`, `authenticate_user`, `create_access_token`, `get_token` and `get_current_active_user`. Tokens use HS256 and expire after 30 minutes.

`gemini_utils.py` — loads the API key, configures the Gemini model, defines the three planner functions (`get_home_recommendations`, `get_party_recommendations`, `get_jewelry_recommendations`), the helper `extract_json_from_response`, the link builder `_add_links`, and the platform dictionaries.

`models.py` — Pydantic classes describing the data that comes in: `RegisterUser`, `UserInDB`, `HomeBudgetInput`, `PartyBudgetInput`, `JewelryBudgetInput`.

`templates/` — `index`, `register`, `login`, `dashboard`, `home_planner`, `party_planner`, `jewelry_planner` and `history` pages.

`static/` — `styles.css` for the look and feel, and the `uploads/` folder for outfit images.